Problem Definition Document

1. Problem Statement

Small businesses and MSMEs in India face significant challenges in managing GST compliance due to manual invoice handling, complex tax rules, and lack of technical knowledge.

There is a need for a simple AI-based solution using OCR that automates invoice processing and simplifies GST filing for non-technical users.

2. Problem Description

Problem Context

In India, GST compliance is a mandatory yet complex process for millions of MSMEs and small businesses. Managing invoices, calculating taxes, and filing returns often involves manual work or dependence on accountants. Many business owners lack proper knowledge of GST rules, leading to frequent errors and missed deadlines. This not only increases the risk of penalties but also adds stress and consumes valuable time. Existing solutions are either too complicated or not affordable for small businesses. As a result, there is a strong need for a simple, automated system that can reduce manual effort and make GST compliance easier and more reliable.

Problem Faced by the Customer

Customers face several key challenges:

- Manual entry of invoices takes a lot of time and effort, especially for businesses with daily transactions
- High chances of human errors in GST calculation and data entry
- Lack of proper understanding of GST rules, tax rates, and filing procedures
- Heavy dependency on accountants, increasing cost and reducing control
- Difficulty in tracking multiple invoices and maintaining proper records
- Missing filing deadlines due to poor reminders or awareness .
- Existing software solutions are often complex, expensive, or not user-friendly for small business owners.

From customer discussions, it was observed that most small business owners find GST compliance confusing and time-consuming. Many of them rely heavily on accountants and still face errors or delays in filing. Customers expressed a strong need for a simple, automated solution that reduces manual work and ensures accuracy.

Typical Customers and Why They Face This Problem

The typical customers include**:**

- Small shop owners (kirana, clothing, electronics)
- Startup founders / small business owners
- Service providers (cafes, salons, repair shops)

They face this problem because:

- Lack of proper knowledge about GST rules and filing process
- No dedicated accounting or finance support
- Limited time to manage compliance along with business work
- Dependence on manual methods or basic tools (like Excel)

Business owners and freelancers confirmed that GST compliance is often confusing and time-consuming, requiring frequent assistance from accountants, indicating that this is a widespread issue among MSMEs.

Customer Validation and Willingness to Pay

During customer interviews:

- Customers said GST filing is confusing and time-consuming
- Many rely on accountants but still face errors and delays
- They expressed frustration with manual work and fear of penalties
- Customers showed strong interest in an automated solution

When asked about a potential AI-based solution:

- Customers said it would save time and reduce manual work
- They felt it would help minimize errors in GST calculation and filing
- Many liked the idea of having a simple and guided system
- They showed interest in using it if it is easy to use and affordable

3. Ideal Customer Profile (ICP)

Overview

The Ideal Customer Profile includes small business owners, freelancers, and MSMEs who regularly deal with GST compliance but struggle with manual processes and lack of knowledge. These users need a simple, automated solution to manage invoices, calculate GST accurately, and file returns without errors. They value ease of use, time-saving features, and affordability, making them highly likely to adopt an AI-based GST compliance system.

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1. **Demographic Characteristics**

- Age Group: 25–55 years
- Occupation: Small business owners, freelancers, shopkeepers
- Location: Urban and semi-urban areas in India
- Income Level: Low to middle income business category
- Business Type: Retail shops, service providers, startups
- GST Status: Registered taxpayers under GST

These users are more likely to require regular GST compliance and often face challenges due to lack of proper knowledge, limited resources, and dependence on manual processes.

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2. Behavioral Characteristics

- Prefer assistance when dealing with GST filing and compliance
- Often rely on accountants or agents for managing GST returns
- Avoid complicated software due to lack of technical knowledge
- Seek faster and error-free solutions for tax filing
- Value convenience and time-saving tools

From interviews, it was observed that most business owners depend heavily on accountants for GST-related tasks, indicating a strong need for a simple and guided automated system.

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3. Psychographic Characteristics

- Lack confidence in handling GST rules and calculations
- Fear penalties or notices due to errors in filing
- Prefer simple, guided, and automated solutions
- Trust tools recommended by accountants or other business owners

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4. Key Pain Points

- Difficulty understanding GST rules and filing procedures
- Errors in GST calculation and invoice entry
- Time-consuming manual processes for managing invoices and returns
- Dependence on accountants or third parties
- Lack of proper tracking and visibility of GST data.

These pain points clearly reflect the common issues faced by customers during GST compliance, as observed from the interviews.

5. Needs and Expectations

- Step-by-step guidance for GST filing
- Automated invoice scanning and data entry (OCR)

- Accurate GST calculation and validation
- Reduced time and effort in compliance
- Secure and reliable handling of financial data

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6. Buying Behavior & Willingness to Pay

- Willing to pay for solutions that:
  o Save time
  o Reduce errors
  o Simplify GST compliance
- Prefer affordable and easy-to-use digital solutions
- More likely to adopt if recommended by accountants or other business owners

Both business owners and accountants acknowledged the usefulness of an AI-based GST solution and showed strong interest in adopting such a system.

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7. Secondary Customers (Institutional ICP)

- Accountants and CA firms
- GST consultants and tax advisors
- Small business service platforms

These stakeholders benefit by reducing manual workload and improving efficiency in GST compliance and filing processes.

Conclusion

The Ideal Customer Profile consists of small business owners, freelancers, and MSMEs, especially those with limited technical and GST knowledge, who face difficulties in managing compliance and filing processes. Their strong need for simplicity, accuracy, and time-saving solutions makes them highly suitable users for the proposed AI-based GST compliance system.

4. Customer Persona for your product

Persona Name: Suresh Reddy

Basic Information

- Age: 41 years
- Gender: Male
- Occupation: Small Restaurant Owner
- Location: Semi-urban area
- Income Level: Middle-class
- Family Status: Married, with one child

- Business Type: Runs a small breakfast restaurant (tiffin center)
- GST Status: Registered under GST

Background

Suresh Reddy runs a small breakfast restaurant where he serves items like idli, dosa, and tea. He manages daily operations such as cooking, billing, and customer service, leaving him with very little time for GST-related work. Although his business is registered under GST, he struggles with maintaining proper invoices and filing returns due to lack of knowledge. He depends on an accountant, but still faces confusion, delays, and occasional errors in GST compliance.

Goals

- Ensure accurate and error-free GST filing
- Avoid penalties by meeting deadlines consistently
- Simplify and speed up the entire compliance process
- Gain independence from accountants and manual work

Challenges / Pain Points

- Does not understand complex GST rules and filing procedures
- Finds tax calculations and invoice management confusing
- Makes mistakes while entering invoice data and filing returns
- Depends on accountants or others for GST compliance
- Feels stressed during GST filing deadlines
- Experiences delays in return submission

From interviews, it was found that many small business owners like Suresh struggle to manage GST tasks on their own due to confusion and lack of proper guidance.

Behavior & Attitude

- Prefers intuitive, step-by-step guided solutions
- Values accuracy, trust, and consistency in financial tasks
- Tends to avoid complicated GST procedures due to fear of errors
- Willing to adopt technology that is easy, reliable, and time-saving

Business owners consistently expressed frustration over unclear GST processes and frequent errors, highlighting a strong inclination toward simple, guided, and dependable solutions—clearly reflecting Suresh's behavioral pattern.

Needs & Expectations

- A simple and user-friendly GST management system

- Step-by-step guidance for GST filing and compliance
- Automatic invoice scanning and data entry to reduce errors
- Quick and accurate GST calculation and return filing
- Assurance of error-free compliance and timely submissions

Motivation to Use the Product

Suresh is motivated to use an AI-based solution because it:

- Saves valuable time in managing GST tasks
- Reduces confusion and dependency on manual processes
- Minimizes errors in calculations and filing
- Simplifies a complex and stressful compliance process

Willingness to Pay

The customer is willing to pay for a solution if:

- It is affordable and cost-effective
- It ensures accurate and faster GST filing
- It reduces effort, confusion, and manual work

Accountants and business owners also suggested that such a solution would be highly beneficial for MSMEs, increasing trust and encouraging adoption.

Quote (Persona Insight)

"I just want a simple way to manage my GST without errors and without depending on others."

Conclusion

Suresh Reddy, a small breakfast restaurant owner, represents a large group of MSMEs who struggle with GST compliance due to complex procedures and limited time. Managing daily operations leaves him with little capacity to handle invoices and filings efficiently. Designing a solution tailored to his needs can simplify compliance, reduce errors, and drive strong real-world adoption among similar small business owners.